

THE UNIVERSITY OF NEWCASTLE
DISCIPLINE OF AEROSPACE SYSTEMS, MECHANICAL AND MECHATRONICS
ENGINEERING

2026

MECH2110 Mechanical Engineering Design 1

Warman Final Report

20% of Course Assessment

Due: 28 April 2026, 5 pm

1 Background

Each year, the Warman Design and Build Competition (Warman Competition) sets a task to be achieved autonomously by a device built by students in their first engineering design course/subject/unit in an Australasian (or other countries, by arrangement) mechanical or mechatronics-based BE or 3+2 ME programme. The task is described in detail by a set of rules released by the Warman Competition committee, which vary from year to year. The rules outline the objective of this year's task, the layout for how/where the task is to be completed, the constraints that must be adhered to, and the methods for earning points to grade the different student devices.

Newcastle University has a proud (and often successful) history of competing in the Warman Competition annually.

As part of MECH2110, students are required to INDIVIDUALLY produce an engineering report detailing the process their team went through when attempting to complete this year's Warman Competition. This report is to cover the entire process your group went through, including (but not limited to) task identification and scoping, initial concepts, concept selection, component specification and selection, subsystem design and testing, prototype design and manufacture, prototype evaluation, and risk management.

Warman Final Report- (20% of Course Assessment)

Submit a copy of your individual report to the link on Canvas. You may appropriately reference your group's conceptual design report.

2 Task

Your task is to produce an engineering report detailing the final design produced of your Warman device. Submit a copy of your individual report to the link on Canvas. You may appropriately reference your group's conceptual design report.

2.1 Report Requirements

The report must have no more than **20 pages of content** (Title page, Table of Contents, List of Figures and Tables, and Appendices are not included). It is suggested that you provide:

1. **Introduction:** Aims and overview of the task.
2. **Final Solution:** A detailed description of the final solution produced by the team, including CAD images showing key functions.
3. **Components and Hardware:** An overview of selection and justification.
4. **Manufacturing:** Description of the methods used to produce the device and why.
5. **Risk Management:** A risk assessment and management plan for device operation.
6. **Testing and Evaluation:** Overview of testing, operation, and an evaluation of performance.
7. **Self-Reflection:** A reflection of the project as a whole and what you would do differently next time.
8. **Conclusion:** Summary of key achievements and project outcomes.

2.2 Professional Standards

All solutions must adhere to the official Warman Design and Build Competition Rules. The report is to be of a professional standard, including (at least):

- Table of contents.
- Correctly labelled and referenced figures and tables.
- Clearly laid out sections and sub-sections.
- Correct referencing of external sources.

Note: Datasheets and other documentation should be added to the appendix and do not count toward the page limit.

Warman Final Report – Marking Schedule

Report Content:	Marks
- Introduction and Task Overview	/5
- Description of Final Solution	/25
- Selection of Components and Hardware	/15
- Manufacturing Description	/15
- Risk Management	/10
- Testing and Evaluation	/10
- Project Self-Reflection	/10
- Report Conclusion	/10
Content Subtotal (C)	/100
Report Presentation:	Factor
Presentation Scalar (P)	0.0 – 1.0
Total Score = C × P	/100

**Note: Presentation serves as a scale factor applied to the total content score.*